



Low-Cost QUADRAPHONIC AMPLIFIER

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Here is a circuit for a simple but useful low-cost quadraphonic (having four channels) amplifier. It can be used for experiments in open or closed spaces, including small home labs and studios. It can also be used for the creation of audio effects and as a small audio-distribution amplifier.

Such an amplifier can be used in mono-, stereo-, quadro-phony, or for audio effects, mixing audio signals, or as small public announcement systems, small audio distribution systems, low-power amplifiers for electrical instruments, noise distribution systems, audio line drivers, bridged and paralleled audio amplifiers, and so on.

For these and similar purposes, we need a multi-channel audio amplifier. In many cases the required power per channel is in the range 0.3W to 1W. Also, preferably, the audio amplifier should be capable of working on a battery, car accumulator or simple 1A to 2A wall adaptor.

To keep this quadraphonic amplifier using LM386 simple and inexpensive, it does not have tone-control potentiometers.

Circuit and working

Circuit diagram of the quadraphonic amplifier is shown in Fig. 1. Circuits of all its four channels are identical. These have separate inputs (CON1, CON2, CON3 and CON4) and separate outputs (CON5, CON6, CON7 and CON8), which can be used for different purposes. Also, the amplifiers can be used in bridge or in parallel mode with appropriate input signals.

Inputs of the four amplifiers have separate volume-control

potmeters VR1, VR2, VR3 and VR4, respectively. Input capacitors C1, C9, C17 and C25 should be preferably not polarised, which makes the circuit independent of the polarity of the DC component of input signals. Capacitors C2, C10, C18 and C26 reduce the sensitivity of amplifiers to high-frequency RFI and EMI cap-

tured by the inputs.

Capacitors C3, C11, C19 and C27 reduce the sensitivity of amplifiers to noise from the power supply. The external power supply connected to CON9 is common for all four channels. The circuit has common fuse F1 and common large filtering capacitor C33 = 1000µF/25V. A well-regulated, low-noise external power supply should be used.

The circuit is built around the popular low-voltage audio-power amplifier LM386. LM386, AZ386, JRC386, NJM386 and similar integrated circuits (ICs) are easily available in the market. These are low-cost and suitable for many audio and control applications.

LM386 is produced by National Semiconductor Corp. in several versions with different power supply voltage ranges. Here, ICs in dual-inline (DIP or DIL) packages are used.

It is important to respect the power dissipation of LM386. In all cases, it is recommended not to go above 80 per cent of the values in the data sheet. This is because everything cannot be taken into consideration at the design stage, and the IC should not operate for a long time at its maximum power dissipation.

LM386N-1 and LM386N-3 have an operating supply range of four volts to 12 volts (maximum 15 volts) and LM386N-4 has an operating supply range of five volts to 18 volts (maximum 22 volts).

Maximum power dissipation of DIP/DIL at +25°C is 1.25W. Table I gives the guaranteed level of the sinusoidal output power (Pout) provided by LM386.

In the datasheet, power dissipation of LM386 is guaranteed on loads

PARTS LIST

Semiconductors:

- IC1-IC4 - LM386 power amplifier
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1, R5, R9, R13 - 1-kilo-ohm
- R2, R6, R10, R14 - 10-ohm
- R3, R7, R11, R15 - 180-ohm
- R4, R8, R12, R16 - 20-ohm
- R17 - 2.2-kilo-ohm
- VR1-VR4 - 10-kilo-ohm preset

Capacitors:

- C1, C4, C9, C12, C17, C20, C25, C28 - 10µF, 25V electrolytic
- C2, C10, C18, C26 - 220pF ceramic
- C3, C11, C19, C27 - 10µF, 35V electrolytic
- C5, C13, C21, C29 - 33nF ceramic disk
- C6, C14, C22, C31, C34 - 100nF ceramic disk
- C7, C15, C23, C30 - 100µF, 25V electrolytic
- C8, C16, C24, C32 - 470µF, 25V electrolytic
- C33 - 1000µF, 25V electrolytic

Miscellaneous:

- CON1-CON4 - 2-pin connector
- CON9 - 2-pin connector terminal
- CON5-CON8 - 3-pin connector terminal
- F1 - 1A fuse
- CON5-CON8 - 8-ohm, 1W loudspeakers
- F1 - Headphones

TABLE I
SINUSOIDAL OUTPUT
POWER PROVIDED
BY LM386

Power supply, + Vee	Load, ohm	Pout minimal, W	Pout typical, W
6V	8-ohm	0.25W	0.325W
9V	8-ohm	0.5W	0.7W
16V (only for LM386N-4)	32-ohm	0.7W	1W